

A Causal Evaluation of Timeouts in the NBA

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Abstract

There is a long-standing belief in the NBA that calling a timeout during an opponent’s scoring run can disrupt their momentum and improve the calling team’s chances of winning. However, the empirical evidence in support of this belief is limited and inconclusive. In this paper, we use play-by-play data from the 2021-2026 NBA seasons to conduct a causal analysis of the effect of timeouts on team performance.

Keywords: Causal inference, sports analytics, NBA, timeouts, momentum, difference-in-differences, synthetic control

1 Significance

The definition and nature of streaks, runs, and momentum itself has been a point of interest and debate for sports fans and analysts for decades. In addition to the papers cited in our related works, there’s a large body of research on the existence and nature of momentum in sports, both on the individual level (e.g. the “hot-hand” effect) and the team level.

On an academic level, the question of whether timeouts disrupt momentum is a interesting case study in causal inference in a complex, dynamic system. It involves issues of selection bias (coaches call timeouts in response to game state), confounding (many factors influence scoring dynamics), and the challenge of defining and measuring momentum itself. The conflicting conclusions in the existing literature suggest that this is a non-trivial question.

The significance of this question extends beyond academic curiosity. If timeouts do have a causal effect on disrupting momentum, it would validate a common coaching strategy and could influence how teams manage their timeouts. There’s also opportunity for further discovery: how does the relationship change conditioned on player substitutions, home-away status, quarter, regular vs post season, team age and maturity, duration and intensity of the run, etc.? There are many more such ablations to consider. Conversely, if timeouts are found to have no effect or even a negative effect, it could lead to a reevaluation of how coaches use their timeouts and potentially change the way teams approach in-game strategy.

2 Related Works

The question of whether timeouts actually disrupt momentum has been studied from several angles, with notably conflicting conclusions. [Assis et al. \(2020\)](#) apply a formal causal

framework to NBA play-by-play data, using matching techniques informed by a causal graph to estimate the counterfactual effect of a timeout. They conclude that timeouts have *no* effect on team performance, and argue that the apparent rebound observed after a timeout is simply regression to the mean: because timeouts tend to be called when the opposing team is scoring at an unusually high rate, any subsequent slowdown reflects the natural tendency of scoring to return to its average rather than a causal effect of the stoppage itself.

Gibbs et al. (2021) revisit the same question but restrict attention to the specific scenario of an opposing team’s scoring run, where the strategic stakes of the timeout decision are highest. Working within the Rubin causal model, they carefully define units under SUTVA and introduce an interpretable outcome based on the score difference to capture broad changes in scoring dynamics. In contrast to Assis et al. (2020), they find that calling a timeout during an opposing run is *slightly disadvantageous* on average, and further show that the magnitude of this effect varies meaningfully by franchise. Their result suggests that comebacks after runs happen frequently enough to create the illusion of timeout efficacy, but that the intervention itself does not help.

Weimer et al. (2023) take a different identification strategy by exploiting mandatory television timeouts, which are triggered at predetermined points in the game and are therefore plausibly exogenous with respect to in-game momentum. Using TV timeouts as a natural experiment that mimics random assignment of a pause, they report an *11.2% decline* in the scoring rate of the team carrying momentum immediately after a TV timeout, and they show this effect is robust to run size, substitutions, and game context. This stands in sharp contrast to the null result of Assis et al. (2020) and the mildly negative result of Gibbs et al. (2021): by removing coach selection bias entirely, Weimer et al. (2023) argue that team-level momentum genuinely exists and that pauses in play do interrupt it.

Qiu et al. (2025) approach the problem from a sports-psychology perspective rather than a causal-inference one, analyzing 4,051 timeouts across 1,235 elite basketball games using linear mixed models to test pre- versus post-timeout momentum over short (48s), medium (96s), and long (144s) windows. They find that timeouts significantly *increase* momentum for teams that are trailing or otherwise disadvantaged, while tending to *decrease* momentum for teams that are already ahead, with contextual factors such as game period and opponent strength substantially moderating the effect. Unlike the three causal papers above, which target an average treatment effect, their results emphasize heterogeneity: timeouts are neither uniformly helpful nor uniformly useless, and their efficacy is highly conditional on game state. Together these four studies motivate our own analysis, which aims to reconcile the tension between the exogenous-timeout evidence for a real momentum effect and the selection-bias-corrected evidence against one.

3 Methods

3.1 Data

We build our analysis on the cdnnba enriched play-by-play data for the 2020–2025 NBA seasons (regular season and playoffs), loaded via our own `CDNNBAMemoPL.load_all()` pipeline. After cleaning, the spine contains 4,183,347 play-by-play rows spanning 7,457 unique games and 1,480,697 possessions, of which 83,662 are coach-called timeouts and 14,561 are inferred mandatory television timeouts.

3.2 Timeout classification

TODO: explain what differs between endogenous and exogenous timeouts? When are teams allowed to substitute players? When are teams likely to alter their strategy? We partition stoppage events into three mutually exclusive groups. **Endogenous** timeouts are coach-initiated stoppages, identified by `actionType == timeout` and `subType` in `{full, challenge}`. These are endogenous in the causal sense because coaches decide when to call them in response to game state. **Exogenous** timeouts are the mandatory television timeouts specified by the NBA rulebook at roughly the 6:59 and 2:59 marks of each period. The `cdnnba` feed does not directly label TV timeouts, so we inject them using a rulebook-based inference procedure and assign them `subType == official_inferred`. We validated this procedure against the explicitly labeled 2013–2016 `nbastatsv3` feed and obtained an F1 of 0.88–0.92, indicating good but imperfect recovery of mandatory timeouts. **Control** events are dead-ball moments with no timeout occurring during an active scoring run; to avoid over-representing long runs, we subsample at most one control per run segment.

3.3 Run and recovery definitions

A *scoring run* at play t is defined as $|\text{streak}(t)| \geq s$, where $\text{streak}(t)$ is the cumulative net score swing within the current scoring segment, computed by resetting the counter whenever the scoring team flips. We refer to the team being outscored as the *suffering team* and the team scoring as the *running team* or alternatively the *streaking team*.

Our primary outcome is the *recovery* metric: the signed lead change from the suffering team’s perspective over a forward window of n minutes. If $\text{lead}(t)$ is the home-minus-away margin at play t , we define

$$\text{recovery}(t, n) = -\text{sign}(\text{streak}(t)) \cdot (\text{lead}(t + n \text{ min}) - \text{lead}(t)),$$

so that positive values mean the suffering team clawed points back during the window. As a secondary outcome we also report `running_team_pts`, the raw points scored by the running team over the forward window, which matches the dependent variable used by [Weimer et al. \(2023\)](#).

3.4 Between-group estimators

As a first pass we compute group means of `recovery` conditional on run size s and forward window n (measured in minutes), and report differences versus the control group using Welch’s t -tests. We sweep $s \in \{4, 5, 6, 7, 8, 10, 12, 15\}$ and $n \in \{0.5, 1, 1.5, 2, 3, 4, 5\}$ minutes, and also stratify by quarter, regular season vs. playoffs, and absolute score margin.

3.5 Within-game matched-twin causal design

TODO: define variables up front. what is suffering margin? the score difference, or the streak size? TODO: break. into two paragraphs. one for the motivation, one for the method. method should be more clear about what qualifies as control and what qualifies as treated

The between-group estimator is vulnerable to game-level confounding: coaches call timeouts in systematically different games than the ones where control events are drawn, and

mandatory TV timeouts likewise cluster in particular game types (TODO which types of games? Is this true?). To remove this bias we use a *within-game matched-twin* design. For each treated event (endogenous or exogenous timeout during a run of size $\geq s$), we search for a control event *in the same game* that also satisfies: the same period; the same sign of **streak** (so the run is against the same team); $|\Delta \text{streak}| \leq 2$ (comparable run magnitude); $|\Delta \text{seconds_remaining}| \leq 120\text{s}$ (comparable clock position within period); and $|\Delta \text{suffering_margin}| \leq 3$ (comparable score state). When multiple candidate controls match, we pick the one minimizing a combined distance over those dimensions (TODO what is the cost coefficient of the weights? Maybe we should include the algorithm here). The per-pair causal estimate is the difference in **recovery** between the treated event and its matched twin, and we aggregate across pairs with a paired *t*-test. This design holds team composition, pace, coaching, and game-level momentum approximately fixed by construction.

3.6 Moderator analyses

To probe heterogeneity we repeat the between-group and matched-twin analyses within slices defined by (i) timeout subtype (**coach_full**, **coach_challenge**, **official_inferred**, **stoppage**); (ii) game phase and clutch status (last five minutes of Q4 with margin ≤ 5); (iii) season-long team quality gap, using each team’s **NET_RATING** from **player_advanced_stats**; (iv) the number of substitutions within a $\pm 30\text{s}$ game-clock window around the event, bucketed as $\{0, 1-2, 3+\}$; and (v) the running team’s in-game cumulative points-per-possession (PPP) at the time of the event, as a proxy for short-term shooting rhythm.

4 Evaluation

4.1 Between-group results

The simple between-group comparison essentially reproduces the weak-to-null effects reported in the prior literature. Across run sizes $s \in \{5, \dots, 10\}$ and forward windows $n \in \{1, \dots, 5\}$ minutes, the endogenous-vs-control gap in **recovery** is never statistically significant (all $|\Delta| < 0.1$ pts, $p > 0.05$ in 15 of 16 cells). Exogenous timeouts, by contrast, look significantly *negative*, with $\Delta \approx -0.25$ to -0.45 pts at the 3-minute window depending on run size ($p < 0.001$). Sweeping the forward window at fixed run size $s = 6$ shows the exogenous penalty growing from -0.20 at 30 seconds to a plateau around -0.27 at 3 minutes, while the endogenous gap hovers near zero throughout. At face value this matches the “timeouts don’t help (and mandatory TV breaks hurt)” consensus. As a secondary check we also computed **running_team_pts** in the style of [Weimer et al. \(2023\)](#) and found a *positive* exogenous-minus-control gap of $+2.8\%$ to $+14.6\%$ depending on window – the opposite sign from Weimer’s reported -11.2% . This is internally consistent with our suffering-team analysis (if the suffering team recovers less, the running team must score relatively more) and likely reflects the post-2017 mandatory-timeout rule changes together with the inferred-rather-than-labeled nature of our TV timeouts.

To avoid confusion we’ll restate our conclusion for this result; the coaches timeouts appear to perform worse than random stoppages of play and exogenous (TV) timeouts. We

believe that this is **not** a causal result, but instead a consequence of the adverse situations that coaches predominantly (TODO correct spelling?) call timeouts during.

4.2 Within-game matched-twin results

TODO clarify what we’re pairing – is the control a non-stoppage point of play? Are we making sure we don’t doubly-count control or test datapoints? j– the answer should go into section 3.5 and linked here

The matched-twin design, which is our cleanest causal test, reverses the headline conclusion of the between-group analysis. With $s = 6$ and a 3-minute window we obtain 8,906 matched endogenous pairs and 2,667 matched exogenous pairs. The paired mean differences are **+0.397 pts for endogenous timeouts** ($t = 27.9$, $p < 0.0001$) and **+0.455 pts for exogenous timeouts** ($t = 11.8$, $p < 0.0001$). Both effects are highly significant and both favor the suffering team; the exogenous effect is in fact slightly *larger* than the endogenous effect, which is consistent with coaches selectively burning timeouts in especially adverse game states where recovery is hard even with help (TODO hyperlink to Future Work: improving the pairing). Breaking the treated population into fine subtypes, every subtype shows the same sign and roughly the same magnitude: `coach_full` +0.396 ***, `coach_challenge` +0.714 * (small sample, $n = 35$), `official_inferred` +0.449 ***, and generic `stoppage` events +0.458 ***. The reversal relative to the between-group results is driven by game-level selection: coach timeouts are concentrated in low-recovery (TODO: what does low-recovery mean?) games, so comparing the endogenous mean to a global control mean absorbs the selection into the baseline and erases the signal. Matching within game cancels this bias by construction, and a simpler within-game mean comparison gives a consistent +0.35 pts effect for endogenous timeouts with the exogenous effect shrinking to near zero (TODO: what does this mean? the exogenous impact minus the endogenous impact is zero?).

We also offer a hypothesis for the exaggerated impact that coaching challenges demonstrate in the data. A successful coaching challenge offers the calling team a break in play, a psychological boost, and tangible in-game benefits (e.g. Free Throws, Possession, etc). If we assume that coaches are skilled at challenging calls, it would follow to reason that this category of game stoppage would show larger effects. The assumption on successful coaching challenges is intuitive: coaches are backed by large analytics and rules experts teams, and the finite number of challenges (1 per game, with a second awarded if the first challenge is successful) means that coaches must use them judiciously. (TODO: re-run notebooks and figure out how the fail / success split looks)

4.3 Mechanism and moderators

Once we have a signal, we can ask where it comes from. Stratifying by the number of substitutions in a ± 30 s window around the event reveals that the endogenous effect is concentrated entirely in timeouts accompanied by personnel changes: with 0 substitutions the endogenous-control gap is -0.05 (n.s.), with 1–2 substitutions it is $+0.10$ (n.s.), and with 3 or more substitutions it rises to $+0.21$ ($p < 0.001$). (TODO: this is confusing. why are all the values less than the average impact cited above?). The exogenous penalty is correspondingly largest (-0.39 , $p < 0.001$) when no subs occur and vanishes when many do

(TODO: again I’m confused by this large disparity with the section above. We need to dig into the data and resolve). This directly confirms the mechanism proposed by [Weimer et al. \(2023\)](#): the value of a timeout is mediated by substitutions rather than by the pause itself. Conditioning on team quality via season-long `NET_RATING`, we find that the endogenous benefit is strongest in evenly matched games (+0.23 pts, $p < 0.05$) and washes out in lopsided matchups where the talent gap dominates the trajectory. Finally, stratifying by the running team’s in-game PPP we find that when the opponent is shooting unusually efficiently (PPP ≥ 1.30), both effects are amplified and point in opposite directions: endogenous timeouts produce +0.24 pts of extra recovery ($p < 0.05$) while exogenous timeouts produce -0.62 pts ($p < 0.001$). This is the clearest evidence in our data of a genuine momentum effect – a coach timeout with substitutions disrupts a hot-shooting opponent, while a pure TV break effectively gives that opponent a rest and lets the run continue.

4.4 Summary

Our headline finding is that timeouts do causally improve the suffering team’s short-term recovery – by roughly +0.4 points over the following three minutes – and that this effect is masked in between-group analyses by a subtle but severe game-level selection bias. The published consensus of “no effect” or “slightly negative effect” ([Assis et al., 2020](#); [Gibbs et al., 2021](#)) reproduces in our data only when we use a between-group estimator; the within-game matched-twin estimator, which holds game identity approximately fixed by construction, produces a strongly positive and highly significant effect for both endogenous and exogenous stoppages. The benefit is overwhelmingly mediated by substitutions (in agreement with [Weimer et al. \(2023\)](#)) and is largest in evenly matched games and against hot-shooting opponents, which matches both coaching intuition and the conditional-momentum findings of [Qiu et al. \(2025\)](#). TODO: update summary once new experiments are done

Future Work

TODO better pairing mechanism. How do you score the quality of a pairing? What is the sweetspot between pair similarity & sample size? TODO metric other than recovery. What about just points per possession / points per minute of the streaking team? What about impact on WP? TODO attempt to build a CGM from the findings TODO measure the impact on the run when the streaking team calls the timeout. Do they substitute players at a greater rate than normal? what mechanisms can we guess at, to explain the outcome TODO Look at more game-break types: official reviews, injuries, ejections, etc. Do they show the same pattern as TV timeouts? Small perturbations in the environment help a lot when reasoning causally

Further experimentation and code changes (TODO delete section once complete)

- “how does the timeout impact for coaching challenges look if we split by successful / failed challenges? add the code to the .py file, and invoke from the notebook (place the call in the relevant section)”

- “there’s also a chance that successful challenges aren’t marked as timeouts. take another look at the event types and check my idea. I think it would also be good to look at the difference in real-world-time (ie 9:43:38.2 pm - 9:47:11.9 PM) to indicate stoppages in play (in case that reveals where the successful challenges might be hiding). Also I thought that there were 35 matched outcomes (according to the paper) - why do you only see 19 now??”
- use real-world-elapsed-time as a feature in TV timeout injection model (retrain and evaluate on nbastatsv3)
- reconcile values between section 4.2 and 4.3
- ensure that we’re only including timeouts called by the suffering team in the endogenous filter
- analyze real-world-elapsed-time as a mechanism
- team age as a moderator (use average age of players on the court at the time of the timeout). Also use the delta between avg age of players on the court before and after the break
- Should we be more selective about including datapoints at the end of the quarters? i.e. never include datapoints in the last k (such as 2) minutes of the quarter, and don’t include datapoints if $t + n$ exceeds the length of the quarter? ie with 3.5 minutes left, include the 3 minute recovery value, but exclude the 4 minute window from analysis
- update NBA data? (I think this would be expensive and tricky because Ted’s data is impossible to download)
- Can we investigate simple substitutions w/out official timeouts? Such as free throws. When else are teams allowed to substitute players?
- Re-run the study, but replace minutes with # possessions (make sure this is an even number, to keep possessions per team the same. an edge case is a technical foul, which awards the transgressed team a free possession. this is infrequent enough that it might not matter)
- reconcile with Weimer finding: he states that exogenous breaks cause 11.2 % drop in scoring by the streaking team has. Do we match that?
- reconcile challenges in matched experiment (section 4.2) with E15.
- re-run experiment post-fixes, flag differences, update conclusion if necessary

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N/A

LLM Usage

`claudeopus-4.6` was utilized extensively during the implementation of this research project, and we wanted to acknowledge its contributions.

The author(s) were responsible for project idea generation, sourcing relevant papers and publications, research roadmap, code scaffolding, and report skeleton.

`Opus` assisted in the implementation: data preprocessing and cleaning, feature generation, significance testing, ablation studies, and report completion. All `Opus` work was closely reviewed and cross-examined.

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Appendix

TODO: add full research-summary.md ablation tables here.